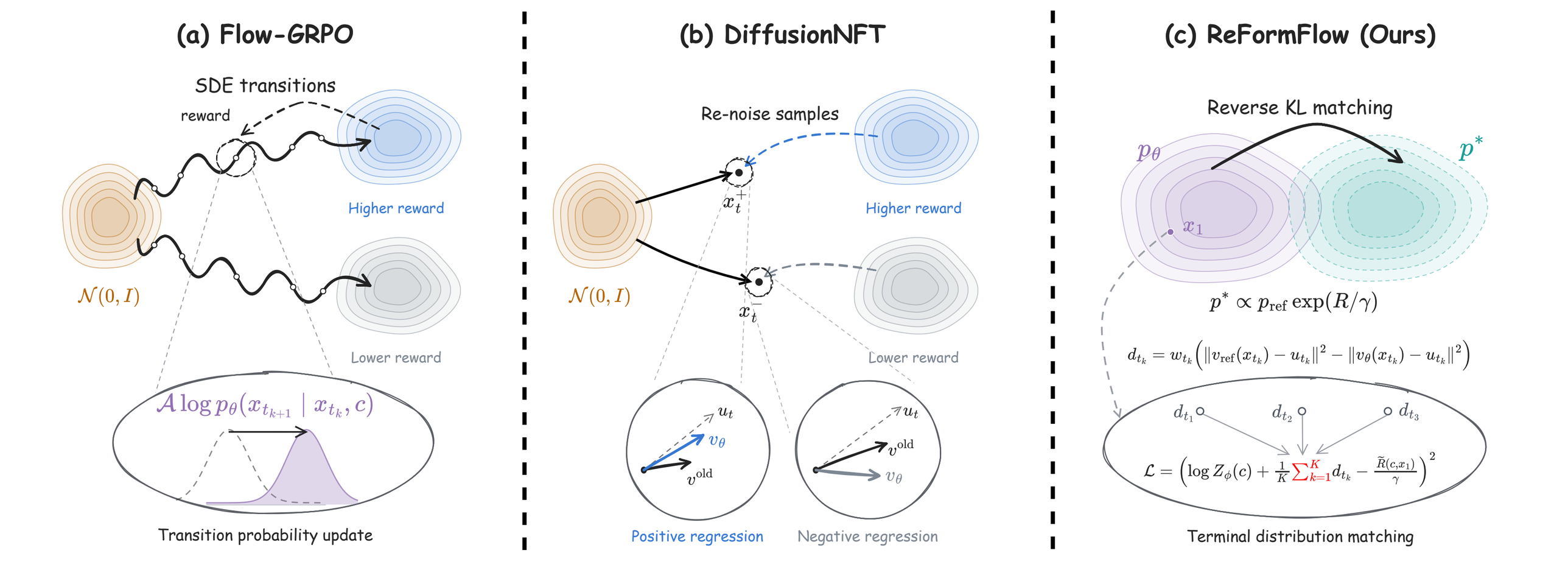

(a) Flow-GRPO
SDE transitions
reward
Higher reward
Lower reward
$\mathcal{N}(0, I)$
$\mathcal{A} \log p_\theta(x_{t_{k+1}} \mid x_{t_k}, c)$
Transition probability update
(b) DiffusionNFT
Re-noise samples
Higher reward
Lower reward
$\mathcal{N}(0, I)$
x_t^+
x_t^-
u_t
v_θ
v^{old}
Positive regression
Negative regression
(c) ReFormFlow (Ours)
Reverse KL matching
p_θ
p^*
x_1
$p^* \propto p_{\text{ref}} \exp(R/\gamma)$
$d_{t_k} = w_{t_k}\left(\|v_{\text{ref}}(x_{t_k}) - u_{t_k}\|^2 - \|v_\theta(x_{t_k}) - u_{t_k}\|^2\right)$
d_{t_1}
d_{t_2}
d_{t_3}
$\mathcal{L} = \left(\log Z_\phi(c) + \frac{1}{K}\sum_{k=1}^{K} d_{t_k} - \frac{\tilde{R}(c,x_1)}{\gamma}\right)^2$
Terminal distribution matching